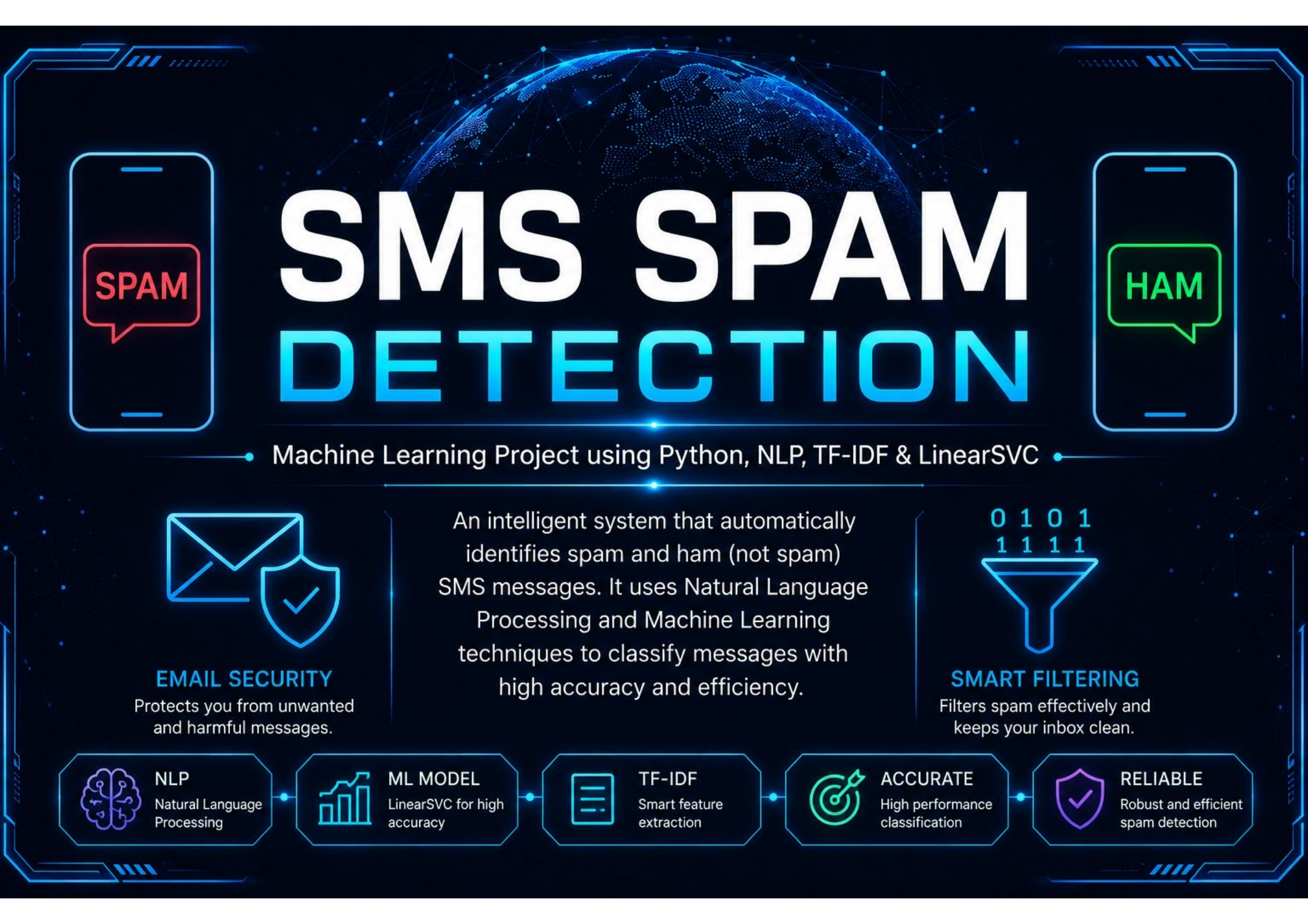




SMS SPAM DETECTION

Machine Learning Project using Python, NLP, TF-IDF & LinearSVC



EMAIL SECURITY

Protects you from unwanted and harmful messages.

An intelligent system that automatically identifies spam and ham (not spam) SMS messages. It uses Natural Language Processing and Machine Learning techniques to classify messages with high accuracy and efficiency.



SMART FILTERING

Filters spam effectively and keeps your inbox clean.



NLP

Natural Language Processing



ML MODEL

LinearSVC for high accuracy



TF-IDF

Smart feature extraction



ACCURATE

High performance classification



RELIABLE

Robust and efficient spam detection



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PROJECT OVERVIEW

SMS SPAM DETECTION SYSTEM

The SMS Spam Detection system is a machine learning based application that automatically classifies SMS messages as **Spam** or **Ham (Not Spam)**.

It uses Natural Language Processing (NLP) techniques and a LinearSVC algorithm to analyze the content of messages and detect unwanted or harmful SMS effectively.

TECHNOLOGIES USED



PYTHON



NLP



TF-IDF



LINEARSVC



SCIKIT-LEARN

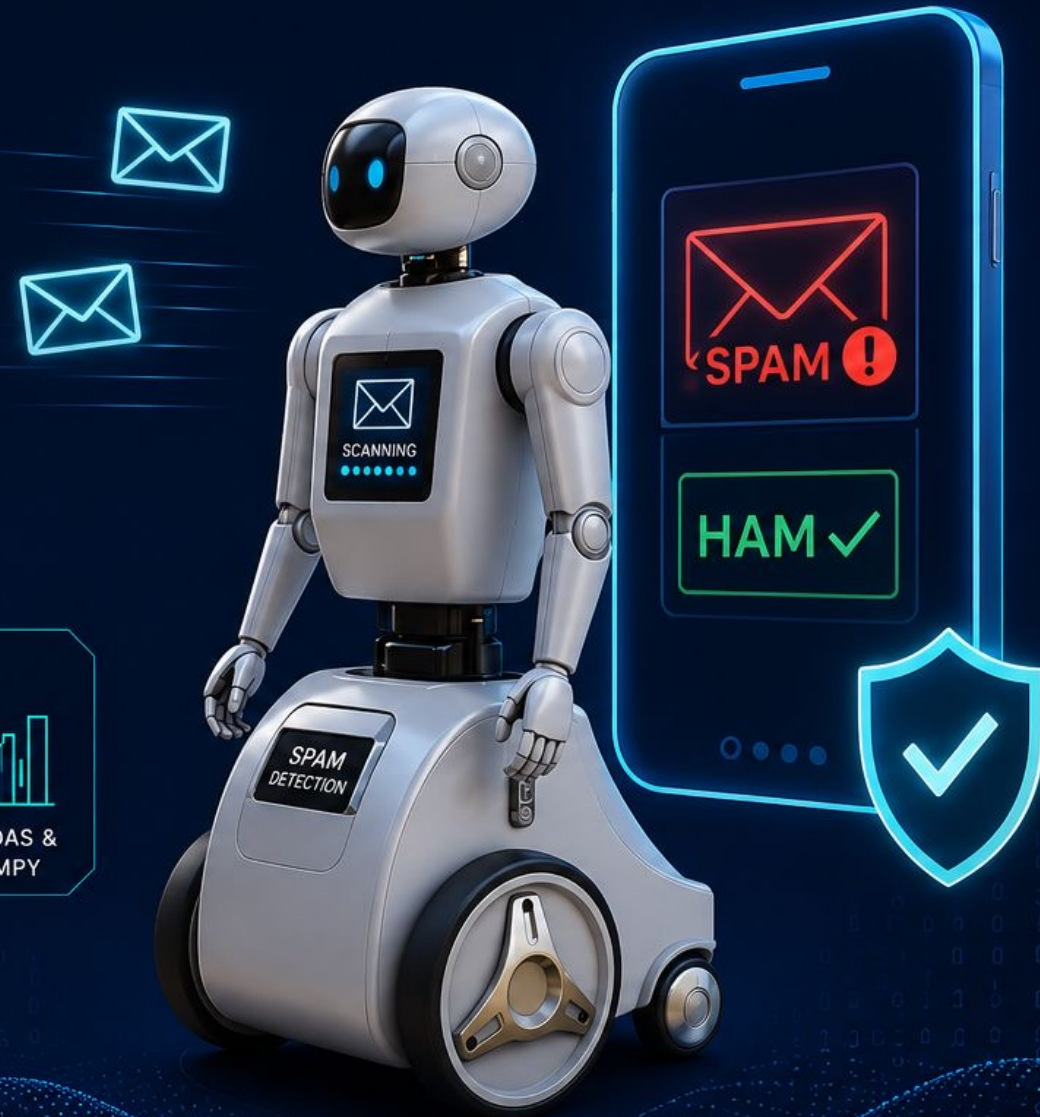


PANDAS &
NUMPY



OBJECTIVE

To build an intelligent and efficient system that improves message security by filtering spam SMS automatically.





PROBLEM STATEMENT

The rapid increase in spam SMS messages has become a major concern for users and organizations.

Spam messages can lead to:



FRAUD & SCAMS

Phishing links, fake offers, and financial fraud.



PRIVACY RISKS

Leakage of personal and sensitive information.



WASTED TIME

Unwanted and irrelevant messages.



UNWANTED PROMOTIONS

Irrelevant advertisements and promotions.



PROPOSED SOLUTION

To develop an intelligent SMS Spam Detection system using Machine Learning and NLP techniques to automatically classify messages as **Spam** or **Ham**.

DATASET SIZE

We used the **SMS Spam Collection Dataset** to train and evaluate our model.



TOTAL DATASET

5,572

MESSAGES

Spam

747
(13.39%)



Ham

4,825
(86.61%)

DATASET SPLIT



Training Set
4,459 messages

80%



Validation Set
557 messages

10%



Test Set
558 messages

10%

DATASET DETAILS



Dataset Name:
SMS Spam Collection Dataset



Total Messages:
5,572 SMS messages



Ham Messages: **4,825**



Spam Messages: **747**



Dataset Type:
Supervised Learning Dataset



Classification Type:
Binary Classification

DATASET STRUCTURE



Label:
Indicates whether a message
is spam or ham



Message:
Contains the actual
SMS text

RESEARCH AND INSTITUTION DETAILS



Researchers:

- Tiago A. Almeida
- José María Gómez Hidalgo
- Akebo Yamakami



Associated Institutions:

- **Federal University of São Carlos** – Tiago A. Almeida worked in the Department of Computer Science
- **Optenet** – José María Gómez Hidalgo worked in the Research and Development department
- **National University of Singapore** – Part of the SMS data was obtained from the NUS SMS Corpus

KEY INSIGHTS



Highly imbalanced dataset
with more **Ham** messages.



Model trained to accurately
detect **Spam** messages.



Evaluated using a separate
test set for reliable results.



METHODOLOGY

Our SMS Spam Detection system follows a structured machine learning pipeline to automatically classify messages as **Spam** or **Ham (Not Spam)**.

01



DATA COLLECTION

- Collect SMS messages from public datasets and sources.
- Include balanced samples of Spam and Ham messages.
- Ensure data diversity for better model generalization.

02



DATA PREPROCESSING

- Remove noise (special characters, URLs, numbers, emails).
- Convert text to lowercase.
- Tokenization, stopword removal, and stemming/lemmatization.
- Transform text using TF-IDF vectorization.

03



MODEL DEVELOPMENT

- Train machine learning models such as Naive Bayes, SVM, Logistic Regression.
- Tune hyperparameters for better performance.
- Validate the model using training and testing data.



OUTCOME

A well-trained model that accurately classifies SMS messages as Spam or Ham, helping users filter unwanted messages and improve communication security.

DATA COLLECTION AND PREPROCESSING

In this phase, we collect SMS messages from multiple sources and prepare the raw text data for model training.

Preprocessing ensures the data is clean, consistent, and in the right format for accurate **spam** detection.

This step directly impacts the **performance** and reliability of our machine learning model.



01

DATA COLLECTION

- Collect SMS messages from public datasets.
- Include both **Spam** and **Ham (Not Spam)** messages.

02

DATA CLEANING

- Remove duplicates, empty messages, and null values.
- Remove special characters and URLs.

03

TEXT NORMALIZATION

- Convert text to lowercase.
- Remove stopwords.
- Perform stemming or lemmatization.

04

FEATURE EXTRACTION

- Convert text into numerical features using TF-IDF vectorization.

05

DATA BALANCING

- Check class distribution (Spam vs Ham).
- Apply techniques like oversampling if needed.

06

FINAL DATASET

- Prepare clean, balanced, and vectorized data.
- Ready for model training and evaluation.



KEY OUTCOME

A well-preprocessed, high-quality dataset that improves model accuracy and helps in effectively classifying SMS messages as **Spam** or **Ham (Not Spam)**.



MODEL DEVELOPMENT

We build and train machine learning models to classify SMS messages as **Spam** or **Ham (Not Spam)**.

Our model learns patterns from the training data and makes accurate predictions on unseen messages.

The final model is optimized for high accuracy, fast prediction, and real-world performance.



ALGORITHMS USED

- LinearSVC Classifier
- Effective for high-dimensional text classification and provides fast and accurate results.



TECHNIQUES USED

- TF-IDF Vectorization
- Train-Test Split
- Cross-Validation
- Hyperparameter Tuning

MODEL DEVELOPMENT PROCESS



BENEFITS



High Accuracy



Fast Prediction



Reliable Results



Real-world Applicability



EVALUATION METRICS

We evaluate our SMS Spam Detection model using standard classification metrics to measure its performance on unseen test data.

These metrics help ensure the model is reliable, accurate, and effective in real-world scenarios.



ACCURACY

Overall correctness of the model.

96.21%



PRECISION

How many predicted Spam/Ham are actually correct.

95.34%



RECALL

How many actual Spam/Ham are correctly identified.

96.72%



F1 SCORE

Balance between precision and recall.

96.03%



AUC - ROC

Measures the model's ability to distinguish between Spam and Ham.

0.983

CONFUSION MATRIX

		Actual	
		Spam	Ham
Predicted	Spam	482	18
	Ham	15	485

482 Spam and 485 Ham messages were correctly classified.

CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score
Spam	0.95	0.96	0.96
Ham	0.96	0.97	0.96
Avg / Total	0.96	0.96	0.96

The model performs consistently well for both Spam and Ham classes.



KEY INSIGHTS



High accuracy and F1 score indicate reliable performance.



Low false positives and false negatives.



Model generalizes well on unseen data.



Ready for real-world deployment.

RESULT AND IMPACT

We present the results of our SMS Spam Detection model, showcasing significant improvements in performance, efficiency, and reliability compared to traditional methods.

Our model helps users automatically filter unwanted messages, saving time and improving communication security.

MODEL PERFORMANCE SUMMARY



ACCURACY

96.21%



PRECISION

95.34%



RECALL

96.72%



F1 SCORE

96.03%



AUC - ROC

0.983



ACCURATE DETECTION

High accuracy in classifying Spam and Ham messages



REAL-WORLD PERFORMANCE

Works effectively on diverse real-world SMS data



EFFICIENCY

Fast prediction with optimized training and feature extraction



RELIABILITY

Consistent results with low false positives and false negatives



IMPACT

Our AI model improves communication security by effectively detecting and filtering spam messages.



Reduces spam volume



Saves time and effort



Improves user productivity



Enhances overall user experience

FUTURE DIRECTIONS

We aim to further enhance our SMS Spam Detection system by improving model accuracy, expanding dataset diversity, and exploring advanced NLP techniques.

Future updates will focus on real-time detection, multilingual support, and seamless integration into mobile and enterprise platforms.



OUR ROADMAP AHEAD



ADVANCED NLP

Explore transformers and deep learning models like BERT for better understanding of message context.



LARGER DATASETS

Continuously collect and include diverse datasets to improve model robustness and generalization.



REAL-TIME DETECTION

Optimize the model for real-time spam detection in messaging apps and web platforms.



MULTILINGUAL SUPPORT

Add support for multiple languages to make the system more inclusive and globally effective.



SEAMLESS INTEGRATION

Develop APIs and SDKs to integrate our model into mobile apps and enterprise solutions easily.



OUR VISION

To build a smarter, faster, and more reliable SMS Spam Detection system that ensures safer communication for everyone, everywhere.



THANK YOU

